Daily Problem Solutions Booklet

Comprehensive Step-by-Step Analytical Solutions

Monday, 17 August 2026

Contents

Integrals

1.1 Medium Integral

Trigonometric Substitution Integral

Evaluate the definite integral:

$$\int_0^1 \sqrt{\frac{2+2\sqrt{1-x^4}}{1-x^4}}\,dx$$

Solution: Let I be the integral. We employ the substitution $x = \sqrt{\sin\theta}$. This gives $x^2 = \sin\theta$, and differentiating yields $2x\,dx = \cos\theta\,d\theta$, so $dx = \frac{\cos\theta}{2\sqrt{\sin\theta}}\,d\theta$. The limits of integration change: when $x = 0$, $\theta = 0$; when $x = 1$, $\theta = \frac{\pi}{2}$. Notice that $1 - x^4 = 1 - \sin^2\theta = \cos^2\theta$. Substituting these into the integral gives:

$$I = \int_0^{\pi/2} \sqrt{\frac{2+2\cos\theta}{\cos^2\theta}} \left(\frac{\cos\theta}{2\sqrt{\sin\theta}}\right) d\theta$$

Using the half-angle identity $2 + 2\cos\theta = 4\cos^2\left(\frac{\theta}{2}\right)$, the square root simplifies to $\frac{2\cos(\theta/2)}{\cos\theta}$ (since $\cos(\theta/2) > 0$ on this interval).

$$I = \int_0^{\pi/2} \frac{2\cos(\theta/2)}{\cos\theta} \cdot \frac{\cos\theta}{2\sqrt{\sin\theta}}\,d\theta = \int_0^{\pi/2} \frac{\cos(\theta/2)}{\sqrt{\sin\theta}}\,d\theta$$

Applying the double-angle identity for sine, $\sin\theta = 2\sin(\theta/2)\cos(\theta/2)$:

$$I = \int_0^{\pi/2} \frac{\cos(\theta/2)}{\sqrt{2\sin(\theta/2)\cos(\theta/2)}}\,d\theta = \frac{1}{\sqrt{2}}\int_0^{\pi/2} \frac{\sqrt{\cos(\theta/2)}}{\sqrt{\sin(\theta/2)}}\,d\theta = \frac{1}{\sqrt{2}}\int_0^{\pi/2} \sqrt{\cot(\theta/2)}\,d\theta$$

Let $u = \frac{\theta}{2}$, then $d\theta = 2\,du$. The bounds become 0 to $\frac{\pi}{4}$:

$$I = \sqrt{2}\int_0^{\pi/4} \sqrt{\cot u}\,du$$

Substituting $v = \sqrt{\cot u}$ leads to a known standard integral that resolves via partial fractions (often related to $\int \sqrt{\tan x}dx$). Solving the resulting rational integral gives:

$$I = \frac{\pi}{2} + \ln(\sqrt{2}+1)$$

The exact value is $\frac{\pi}{2} + \ln(\sqrt{2}+1)$.

1.2 Hard Integral

Inverse Hyperbolic Integral

Evaluate the definite integral:

$$\int_1^\infty \frac{\cosh^{-1}(\sqrt{x})}{\sqrt{x}(x+4)^3}\,dx$$

Solution: Let $u=\sqrt{x}$. Then $du=\frac{1}{2\sqrt{x}}\,dx$, which implies $\frac{dx}{\sqrt{x}}=2\,du$. The bounds remain 1 to ∞. The integral transforms into:

$$I=2\int_1^\infty \frac{\cosh^{-1}(u)}{(u^2+4)^3}\,du$$

To evaluate this, we define a parameter-dependent integral:

$$J(k)=\int_1^\infty \frac{\cosh^{-1}(x)}{x^2+k}\,dx=\frac{\pi}{2\sqrt{k}}\sinh^{-1}(\sqrt{k})$$

We need to obtain the denominator $(x^2+4)^3$. This can be achieved by taking the second derivative of $J(k)$ with respect to k:

$$J''(k)=\frac{d^2}{dk^2}\int_1^\infty \frac{\cosh^{-1}(x)}{x^2+k}\,dx=\int_1^\infty \frac{2\cosh^{-1}(x)}{(x^2+k)^3}\,dx$$

This matches our integral I exactly when evaluated at $k=4$. Differentiating $J(k)=\frac{\pi}{2}k^{-1/2}\ln(\sqrt{k}+\sqrt{k+1})$ twice with respect to k requires careful application of the product and chain rules:

$$J'(k)=\frac{\pi}{4}\left[\frac{1}{k\sqrt{k+1}}-\frac{\sinh^{-1}(\sqrt{k})}{k^{3/2}}\right]$$

$$J''(k)=\frac{\pi}{4}\left[\frac{3}{2k^{5/2}}\sinh^{-1}(\sqrt{k})-\frac{4k+3}{2k^2(k+1)^{3/2}}\right]$$

Now, evaluate this expression at $k=4$:

$$\sqrt{k}=2,\quad k^{5/2}=32,\quad k^2=16,\quad (k+1)^{3/2}=5\sqrt{5},\quad 4k+3=19$$

Substituting these values:

$$I=J''(4)=\frac{\pi}{4}\left[\frac{3}{64}\ln(2+\sqrt{5})-\frac{19}{32(5\sqrt{5})}\right]=\frac{\pi}{4}\left[\frac{3\ln(2+\sqrt{5})}{64}-\frac{19\sqrt{5}}{800}\right]$$

Distributing the $\frac{\pi}{4}$:

$$I=\frac{3\pi\ln(2+\sqrt{5})}{256}-\frac{19\sqrt{5}\pi}{3200}$$

Differentials

2.1 Medium Differential

Orthogonal Trajectories

Find a family of curves α orthogonal to $y = Ce^{x+y}$. Given the initial condition $\alpha(0) = 2$, find x when $y = e^5 + 1$.

Solution: First, we find the differential equation governing the original family of curves. Taking the natural logarithm of both sides:

$$\ln y = \ln C + x + y$$

Differentiating implicitly with respect to x:

$$\frac{1}{y}\frac{dy}{dx} = 1 + \frac{dy}{dx} \implies \frac{dy}{dx}\left(\frac{1}{y} - 1\right) = 1 \implies \frac{dy}{dx} = \frac{y}{1-y}$$

For orthogonal trajectories, the slope must be the negative reciprocal of the original slope. Thus, the differential equation for the family α is:

$$\frac{dy}{dx} = \frac{y-1}{y}$$

This is a separable differential equation:

$$\frac{y}{y-1}\,dy = dx \implies \left(1 + \frac{1}{y-1}\right) dy = dx$$

Integrating both sides yields the family of orthogonal curves α:

$$y + \ln|y-1| = x + K$$

We use the initial condition $\alpha(0) = 2$ (meaning $y = 2$ when $x = 0$) to find the constant K:

$$2 + \ln(2-1) = 0 + K \implies 2 + 0 = K \implies K = 2$$

The exact equation for the specific curve is:

$$x = y + \ln(y-1) - 2$$

We are asked to find x when $y = e^5 + 1$:

$$x = (e^5 + 1) + \ln(e^5 + 1 - 1) - 2 = e^5 + 1 + \ln(e^5) - 2$$

$$x = e^5 + 1 + 5 - 2 = e^5 + 4$$

The value of x is $e^5 + 4$.

2.2 Hard Differential

Integro-Differential Equation

Find the value of $y(1)$ for the continuous function $y(x)$ defined on $[0, \infty)$ that satisfies the non-linear non-autonomous integro-differential equation:

$$y'(x) + y(x) = \int_0^x y(t)y(x-t)\,dt$$

subject to the initial condition:

$$\lim_{x \to 0^+} y(x) = 1$$

Find the exact value of $y(1)$.

Solution: Notice that the right-hand side is the convolution of $y(x)$ with itself, $(y * y)(x)$. We apply the Laplace Transform to both sides. Let $Y(s) = \mathcal{L}\{y(x)\}$. Using the initial condition $y(0) = 1$:

$$(sY(s) - 1) + Y(s) = Y(s)^2$$

Rearranging yields a quadratic equation in terms of $Y(s)$:

$$Y(s)^2 - (s+1)Y(s) + 1 = 0$$

Solving for $Y(s)$ using the quadratic formula:

$$Y(s) = \frac{(s+1) \pm \sqrt{(s+1)^2 - 4}}{2}$$

Since we require $\lim_{s\to\infty} Y(s) = 0$ for a valid Laplace transform of a continuous function, we must choose the negative sign:

$$Y(s) = \frac{(s+1) - \sqrt{(s+1)^2 - 4}}{2}$$

This transform takes the form of a shifted function. Let $F(s) = \frac{s-\sqrt{s^2-4}}{2}$, so that $Y(s) = F(s+1)$. It is a known identity in operational calculus that:

$$\mathcal{L}^{-1}\left\{s - \sqrt{s^2 - a^2}\right\} = \frac{a}{t} I_1(at)$$

Where I_1 is the modified Bessel function of the first kind. Setting $a = 2$:

$$\mathcal{L}^{-1}\left\{\frac{s - \sqrt{s^2 - 4}}{2}\right\} = \frac{1}{2}\left(\frac{2}{t} I_1(2t)\right) = \frac{I_1(2t)}{t}$$

Applying the s-shifting theorem due to the $(s+1)$ term, the inverse Laplace transform is:

$$y(t) = e^{-t}\frac{I_1(2t)}{t}$$

To find $y(1)$, we substitute $t = 1$:

$$y(1) = e^{-1}\frac{I_1(2)}{1} = \frac{I_1(2)}{e}$$

Derivatives

3.1 Medium Derivative

Inverse Trigonometric Minimum

Compute the minimum value of the expression:

$$(\sin^{-1}(x))^2 + (\cos^{-1}(x))^2$$

Solution: Let $f(x) = (\arcsin x)^2 + (\arccos x)^2$. We employ the fundamental inverse trigonometric identity:

$$\arcsin x + \arccos x = \frac{\pi}{2}$$

Let $u = \arcsin x$. Since the domain of x is $[-1, 1]$, the range of u is $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$. Substitute $\arccos x = \frac{\pi}{2} - u$ into the expression:

$$f(u) = u^2 + \left(\frac{\pi}{2} - u\right)^2 = u^2 + \frac{\pi^2}{4} - \pi u + u^2 = 2u^2 - \pi u + \frac{\pi^2}{4}$$

This is a quadratic function in u opening upwards. We complete the square to find its minimum:

$$f(u) = 2\left(u^2 - \frac{\pi}{2}u\right) + \frac{\pi^2}{4} = 2\left(u - \frac{\pi}{4}\right)^2 - 2\left(\frac{\pi^2}{16}\right) + \frac{\pi^2}{4}$$

$$f(u) = 2\left(u - \frac{\pi}{4}\right)^2 - \frac{\pi^2}{8} + \frac{2\pi^2}{8} = 2\left(u - \frac{\pi}{4}\right)^2 + \frac{\pi^2}{8}$$

The minimum value occurs when the squared term is zero, which is at $u = \frac{\pi}{4}$. This corresponds to $\arcsin x = \frac{\pi}{4}$, meaning $x = \frac{1}{\sqrt{2}}$, which is valid within our domain. The minimum value of the expression is exactly $\frac{\pi^2}{8}$.

3.2 Hard Derivative

Gamma Zeta & Dilogarithm

Let $f(x) = [\Gamma(x)]^{\zeta(x)} + \mathrm{Li}_2(2-x)$ for $x > 1$. Find $f'(2)$.

Solution: Let $A(x) = [\Gamma(x)]^{\zeta(x)}$ and $B(x) = \mathrm{Li}_2(2-x)$. First, we find $A'(x)$ using logarithmic differentiation:

$$\ln A(x) = \zeta(x) \ln \Gamma(x)$$

$$\frac{A'(x)}{A(x)} = \zeta'(x) \ln \Gamma(x) + \zeta(x) \frac{\Gamma'(x)}{\Gamma(x)} = \zeta'(x) \ln \Gamma(x) + \zeta(x)\psi(x)$$

where $\psi(x)$ is the digamma function. Evaluate at $x = 2$: $\Gamma(2) = 1$, so $\ln \Gamma(2) = 0$. $A(2) = [\Gamma(2)]^{\zeta(2)} = 1^{\pi^2/6} = 1$. $\zeta(2) = \frac{\pi^2}{6}$. $\psi(2) = H_1 - \gamma = 1 - \gamma$, where γ is the

Euler-Mascheroni constant.

$$A'(2) = 1 \cdot \left[\zeta'(2) \cdot 0 + \frac{\pi^2}{6}(1-\gamma) \right] = \frac{\pi^2}{6}(1-\gamma)$$

Next, we differentiate $B(x) = \mathrm{Li}_2(2-x)$. The derivative of the dilogarithm is $\frac{d}{dz}\mathrm{Li}_2(z) = -\frac{\ln(1-z)}{z}$. Using the chain rule:

$$B'(x) = -\frac{\ln(1-(2-x))}{2-x} \cdot (-1) = \frac{\ln(x-1)}{2-x}$$

Evaluating $B'(2)$ directly yields an indeterminate form $\frac{0}{0}$. We apply L'Hôpital's rule:

$$\lim_{x\to 2} \frac{\ln(x-1)}{2-x} = \lim_{x\to 2} \frac{\frac{1}{x-1}}{-1} = \frac{1}{-1} = -1$$

Thus, $B'(2) = -1$. Combining both derivatives gives:

$$f'(2) = A'(2) + B'(2) = \frac{\pi^2}{6}(1-\gamma) - 1$$

Physics & Engineering

4.1 Mechanics

Oscillating Mass on a Wire

A mass $m = 2$ kg moves along a frictionless horizontal wire. It is attached to a spring with constant $k = 49\ \text{N m}^{-1}$. The fixed end is $H = 4$ m vertically above the wire and the spring has natural length $\ell_0 = 2$ m. If x is measured from the point below the fixed end, find the natural angular frequency ω (in rad s^{-1}) of small horizontal oscillations about $x = 0$.

Solution: Let x be the horizontal displacement. The length of the stretched spring is $L(x) = \sqrt{x^2 + H^2}$. The tension in the spring is $T(x) = k(L(x) - \ell_0) = k(\sqrt{x^2 + H^2} - \ell_0)$. The restoring force is the horizontal component of this tension:

$$F_x = -T(x)\sin\theta = -k(\sqrt{x^2 + H^2} - \ell_0)\frac{x}{\sqrt{x^2 + H^2}} = -kx\left(1 - \frac{\ell_0}{\sqrt{x^2 + H^2}}\right)$$

For small oscillations, we approximate the force by linearizing around $x = 0$. Using the binomial expansion for the denominator:

$$F_x \approx -kx\left(1 - \frac{\ell_0}{H}\right)$$

This takes the form of Hooke's Law, $F_x \approx -k_{\text{eff}}x$, giving an effective spring constant:

$$k_{\text{eff}} = k\left(1 - \frac{\ell_0}{H}\right)$$

Substitute the given values ($k = 49$, $H = 4$, $\ell_0 = 2$):

$$k_{\text{eff}} = 49\left(1 - \frac{2}{4}\right) = 49(0.5) = 24.5\ \text{N m}^{-1}$$

The natural angular frequency ω is:

$$\omega = \sqrt{\frac{k_{\text{eff}}}{m}} = \sqrt{\frac{24.5}{2}} = \sqrt{12.25} = 3.5\ \text{rad s}^{-1}$$

4.2 Quantum Mechanics

Photoelectric Magnetic Orbits

Metal A has work function $\phi_A = 4.00$ eV. Both metals are illuminated by photons of energy 6.00 eV. In the same uniform magnetic field, their maximum photoelectron orbit radii are $r_A = r$ and $r_B = \frac{3r}{2}$. Find the ratio $\frac{\phi_A}{\phi_B}$ to two decimal places.

Solution: Using Einstein's photoelectric equation, the maximum kinetic energy K of emitted electrons is $K = E_{\text{photon}} - \phi$. For Metal A:

$$K_A = 6.00 \text{ eV} - 4.00 \text{ eV} = 2.00 \text{ eV}$$

When a charged particle moves in a uniform magnetic field B, its orbital radius is $r = \frac{\sqrt{2mK}}{eB}$. Since both metals are in the same magnetic field, the radius is directly proportional to the square root of the kinetic energy: $r \propto \sqrt{K}$. We are given $r_B = 1.5 r_A$. Setting up the ratio:

$$\frac{r_B}{r_A} = \sqrt{\frac{K_B}{K_A}} = 1.5 = \frac{3}{2}$$

Squaring both sides gives the kinetic energy ratio:

$$\frac{K_B}{K_A} = \frac{9}{4} = 2.25 \implies K_B = 2.25 \times 2.00 \text{ eV} = 4.50 \text{ eV}$$

Now we find the work function for Metal B:

$$K_B = 6.00 - \phi_B \implies 4.50 = 6.00 - \phi_B \implies \phi_B = 1.50 \text{ eV}$$

The ratio $\frac{\phi_A}{\phi_B}$ is:

$$\frac{\phi_A}{\phi_B} = \frac{4.00}{1.50} = \frac{8}{3} \approx 2.67$$

4.3 Electric Fields

Series Cylindrical Wires

Two cylindrical wires are joined end to end in series. The first has length $L_1 = 2.40$ m, radius $r_1 = 0.50$ mm and resistivity $\rho_1 = 1.10 \times 10^{-6}\,\Omega$ m. The second has length $L_2 = 3.00$ m, radius $r_2 = 0.30$ mm and resistivity $\rho_2 = 1.68 \times 10^{-8}\,\Omega$ m. Find the total resistance in Ω to two decimal places.

Solution: The resistance of a cylindrical wire is given by $R = \rho\frac{L}{A} = \rho\frac{L}{\pi r^2}$. For the first wire: $r_1 = 0.50 \text{ mm} = 0.50 \times 10^{-3}$ m.

$$R_1 = \frac{(1.10 \times 10^{-6})(2.40)}{\pi(0.50 \times 10^{-3})^2} = \frac{2.64 \times 10^{-6}}{\pi(0.25 \times 10^{-6})} = \frac{2.64}{0.25\pi} = \frac{10.56}{\pi}\,\Omega$$

For the second wire: $r_2 = 0.30 \text{ mm} = 0.30 \times 10^{-3}$ m.

$$R_2 = \frac{(1.68 \times 10^{-8})(3.00)}{\pi(0.30 \times 10^{-3})^2} = \frac{5.04 \times 10^{-8}}{\pi(0.09 \times 10^{-6})} = \frac{56 \times 10^{-2}}{\pi} = \frac{0.56}{\pi}\,\Omega$$

Since the wires are in series, the total resistance R is the sum:

$$R = R_1 + R_2 = \frac{10.56}{\pi} + \frac{0.56}{\pi} = \frac{11.12}{\pi}\,\Omega$$

Evaluating this numerically:

$$R \approx \frac{11.12}{3.14159} \approx 3.5396\,\Omega$$

Rounded to two decimal places, the total resistance is $3.54\,\Omega$.